

Stage Motion Control

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Stage Motion Control: Nonlinear Integrators Revisited

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Keywords

frequency-domain design, Lyapunov methods, motion control, nonlinear integrators, stability tools

Abstract

Linear integrators are well known for their ability to counter static forces and improve low-frequency disturbance rejection properties in control systems. However, linear integrators introduce phase lag, which is a frequency-dependent time shift or delay. Since the early introduction of the Clegg integrator, nonlinear integrators have held the promise of providing phase advantages over linear integrators when evaluated from the perspective of a describing function. This could potentially reduce delay and therefore provide the means to surpass linear design limitations; for example, overshoot and settling times can be reduced or even avoided. In addition, loop gains can be increased, which improves the low-frequency disturbance rejection properties. For five nonlinear integrators—the Clegg integrator, a generalized first-order reset integrator called the constant-in-gain-lead-in-phase (CgLp) element, a hybrid integrator–gain system, a variable gain integrator, and a split-path integrator—this article provides a comparison and overview of recent developments in stage motion control. Benchmark examples are taken from the industrial practice of wafer scanners, which form the pivotal machines used in the manufacturing of computer chips.

1. INTRODUCTION

In the semiconductor industry, stage systems perform highly accurate positioning tasks under aggressive motion-tracking profiles (1). To meet the ever-tightening control specifications on throughput and position accuracy, the control designs of these systems rely on both feedforward and feedback control (2). Linear feedback control, however, is known for its inherent design limitations (3, 4). These limitations find their roots in the response time or delay of the feedback controller in counteracting disturbances or reducing error. At the same time, the linear feedback controller contains dynamic components (e.g., integrators), the output of which lags the input, leading to a negative phase shift (or phase lag) that degrades a timely response even further. Control engineers have long known that delay has a negative impact on the stability properties of a feedback system and therefore limits stage systems in achieving high bandwidths for disturbance rejection as well as high tracking performance for position accuracy. Nonlinear feedback control has provided a way to overcome these limitations (5–7). This fueled the motivation for nonlinear control of essentially linear systems—i.e., the idea that a continuous nonlinear, hybrid, or discontinuous switching controller in closed loop may exceed the performance that can be achieved by any time-invariant feedback controller without such switching (8–10).

Early examples of nonlinear feedback control to improve the transient response of a linear servo mechanism were proposed by Lewis (11) in 1953 and by Kalman (12) through nonlinear gain elements in 1955. For integrators, the Clegg integrator (13) from 1958 provides an interesting view on phase lag. By resetting the integrator state upon a zero crossing of its input, the Clegg integrator guarantees the sign equivalence of its input–output pair. As a result, its describing function—the quasi-linear function that by approximation describes the transfer characteristics of a nonlinearity (14)—has the same magnitude characteristics as the linear integrator but with 38.1° of phase lag instead of the 90° corresponding to the linear integrator. This notion of breaking free from Bode's (15) gain–phase relationship was further pursued by Foster et al.'s (16) introduction of the split-path filter in 1966 and Bailey's (17) introduction of the driven limiter, also in 1966. The describing function of the split-path filter showed a constant gain with a phase lead. In 1975, the first-order reset element (FORE) was introduced by Horowitz & Rosenbaum (18) (for a more complete overview, see 6). By introducing a pole, the FORE turned the reset integrator into a weak integrator. In 1977, Karybakas (19) introduced the split-path integrator, which put a gain with a phase lead (i.e., the split-path filter) in series with a linear integrator. Its describing function then shows an integrator with zero phase lag.

Modern nonlinear integrators have obvious links to these early developments. Variable gain integrators (20)—e.g., as discussed by Armstrong et al. (21, 22)—relate to the nonlinear gain ideas of Lewis (11) and Kalman (12). As an interesting example, the phase-based variable gain integrator (23) is a piecewise linear system that switches between static modes. Corrective action takes place only if the system moves away from its desired position. Another example is the hybrid integrator–gain system (HIGS) (24), which is a piecewise linear system that switches between a static and a dynamic mode. In contrast to the Clegg integrator and FORE, which are impulsive systems, HIGS has continuous input–output behavior that results from either processing an input through a linear integrator or multiplying the input with a constant gain. HIGS does have a discontinuous vector field. Discontinuities are considered undesirable in high-precision machinery like wafer scanners because they induce unwanted excitation of lightly damped resonances in the structural dynamics of these machines (25). A HIGS-based integrator is obtained by using a HIGS in combination with a linear filter architecture (see, e.g., 26), which is comparable to the idea of Karybakas (19) in the split-path integrator. Other examples of HIGS-based controllers include the HIGS-based low-pass filter (27) and the HIGS-based proportional–integral–derivative (PID) controller (as discussed in Reference 28 and references therein).

Closely related to HIGS is the constant-in-gain-lead-in-phase (CgLp) element (29). CgLp is a generalized FORE in combination with a linear lead filter that through its describing function gives a constant gain with a phase lead, which was the central idea of Foster et al. (16), as further discussed by Zoss et al. (30) and Aangenent et al. (31). For integration, the CgLp element, like HIGS, is generally used within a linear PID-based control architecture (32). As CgLp is based on state resets, it is an impulsive system. Several examples of CgLp-based controllers are discussed in Reference 33 and references therein.

As a last example, the reintroduced split-path integrator (34) is based largely on the original ideas of Foster et al. (16). The difference between this split-path integrator and Karybakas's (19) is mainly in the location of the linear integrator, which is brought before the rectifier (34) and makes the differentiator obsolete. To deal with noisy inputs, input filtering has been proposed in the form of the filtered split-path integrator (35, 36).

Nonlinear integrators possess several useful properties for control design. First, the input-output relation of the Clegg integrator, the phase-based variable gain element, the HIGS element, the CgLp element, and the split-path filter show sign equivalence. Sign equivalence translates via describing function analysis to phase advantages in the frequency domain, which control engineers can use to design for increased controller bandwidth and low-frequency disturbance suppression. Second, nonlinear integrators can inherently outperform linear integrators. For a linear time-invariant (LTI) closed-loop system with a plant that has a real pole in the right half-plane, any stabilizing (linear) controller induces a step response that must overshoot by an amount proportional to the value of that pole (4). For the phase-based variable gain integrator in Reference 23, reset control system in References 37 and 38, and HIGS-based integrator in Reference 39, examples are shown of nonlinear closed-loop systems under the aforementioned conditions that do not induce any overshoot, thereby surpassing this inherent limitation of the linear control design. Third, all considered nonlinear integrators have homogeneity properties, i.e., scaling invariance. Homogeneity is considered a useful property in the nonlinear control design of motion systems as an enabler of loop shaping. It leads to a unique frequency response description, as obtained from describing function analysis. Other properties, like well-posedness in the sense of existence and forward completeness of solutions for nonlinear integrators, have been addressed, for example, by Baños & Barreiro (40) for reset control systems and by Deenen et al. (41) and Sharif et al. (42) for HIGS. For HIGS, the connection to projected dynamical systems and extensions thereof (43) pave the way for notions like uniqueness of solutions and incremental stability (44, 45).

Over the past few decades, significant progress has also been made in the stability and performance analysis of nonlinear control systems (46). A clear example is given by the $\mathcal{H}_\beta$ condition in Reference 47, which is a strictly positive real condition that is necessary and sufficient for the existence of a common quadratic Lyapunov function. The $\mathcal{H}_\beta$ condition offered the first formal proof of closed-loop stability of reset (or impulsive) control systems (40). It allows for both a frequency-domain graphical check and a test by linear matrix inequalities (LMIs) and is equivalent to the output strict passivity condition (48). For continuous piecewise linear integrators, like the variable gain integrator or HIGS, a common quadratic Lyapunov function is often found by application of the circle criterion (49, 50). Finding a common quadratic Lyapunov function, however, in many cases yields a conservative assessment of closed-loop stability. Less conservative results have been reported with piecewise quadratic Lyapunov functions of HIGS-based systems (41). The search for these functions is formulated as convex optimization problems in terms of numerically tractable LMIs for continuous time (51) as well as discrete time (52). Regarding performance analysis, upper bounds on the $\mathcal{L}_2$ gain and $\mathcal{H}_2$ norm have been formulated (53). Van den Eijnden et al. (54) discussed frequency-domain results that extend classical results like the circle and Popov criteria (55, 56) and at the same time enable graphical checks using (measured)

frequency response data; they also make explicit use of relaxation terms to increase the admissible class of Lyapunov functions certifying stability.

The main objective of this article is to provide control engineers with an overview of nonlinear integrators, their properties, and when to use them. For this purpose, we assess five nonlinear integrators: the Clegg integrator, a CgLP integrator, the HIGS integrator, a variable gain integrator, and a split-path integrator. The key questions to be answered are what design freedom each nonlinear integrator brings with respect to a linear integrator and how this freedom can be used to enhance performance.

The remainder of the article is organized as follows. First, Section 2 highlights the problem context in which nonlinear integrators have proven to be highly beneficial, namely the nonlinear motion control of high-accuracy and fast-moving stage systems. Second, in Section 3, properties of the five nonlinear integrators are discussed in the time domain as well as in the frequency domain. Third, conceptual tools for closed-loop stability analysis are presented in Section 4; this covers both time-domain tests formulated using LMIs and graphical tests based on frequency response data. Fourth, Section 5 presents two benchmark examples: a step response benchmark based on a plug-and-play comparison and a disturbance rejection benchmark based on a carefully designed stage disturbance spectrum that uses optimized controller settings. Both examples are taken from a prototype wafer stage system. Finally, Section 6 summarizes the main conclusions and recommendations for further research.

2. THE STAGE CONTROL PROBLEM

To explain the stage control context, consider the block scheme shown in **Figure 1a**, where $\mathcal{P}$ is the stage system, C_{fb} is the feedback controller, and C_{ff} is the feedforward controller. A physical representation of a (wafer) stage system is shown in **Figure 1b**, where the motion of a platform (or stage) in the scanning y direction is conducted via a linear guidance and relative to a base. For ease of notation, all systems are considered to be single-input, single-output. Reference trajectory r , which is a priori known (57); input disturbance d_u ; output disturbance d_y ; and output noise n_y are considered exogenous inputs.

Stage systems are found in wafer scanners, which are the machines used to manufacture computer chips that are based on the principles of photolithography (58, 59). In wafer scanners, an (extreme) ultraviolet light beam is used to capture an image taken from exposing a blueprint of a chip on a mask called a reticle; the image is then scaled by a factor of four using a lens or mirror system and projected onto the light-sensitive layers of a 300-mm silicon wafer. In wafer scanning,

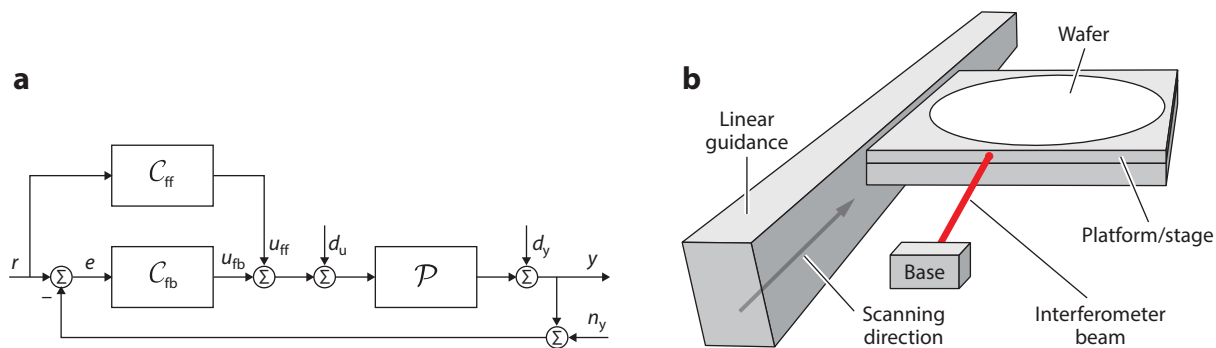


Figure 1

(a) Control block scheme. (b) Wafer stage system.

the stage systems (i.e., the reticle stage supporting the reticle and the wafer stage supporting the wafer) conduct scanning motions that require nanometer tracking accuracy while being subjected to several g 's of acceleration. These accelerations are needed to meet throughput specifications that can be as high as 400 wafers per hour. Stage systems in wafer scanners can be viewed as floating mass systems that in the absence of deformation are described by double integrator behavior. More precisely, the transfer function $\mathcal{P}(s)$ of a wafer stage system can be represented as

$$\mathcal{P}(s) = \underbrace{\frac{1}{ms^2}}_{\mathcal{P}_{\text{rb}}(s)} - \underbrace{\sum_{i=2}^n \frac{c_i}{s^2 + 2\zeta_i\omega_i s + \omega_i^2}}_{\mathcal{P}_{\text{nr}}(s)} - \underbrace{\sum_{i=n+1}^r \frac{c_i}{\omega_i^2}}_{\mathcal{P}_{\text{res}}}, \quad 1.$$

where m is the mass of the stage. In this transfer, the rigid body part of the stage is denoted by $\mathcal{P}_{\text{rb}}(s)$. The nonrigid body part is denoted by $\mathcal{P}_{\text{nr}}(s)$ and contains the dimensionless damping ζ_i , the natural frequency ω_i , and the mode scaling factor c_i corresponding to the i th nonrigid body mode. In total, Equation 1 covers $r > n$ modes, of which $n - 1 > 0$ nonrigid body modes are fully taken into account. For the remaining $r - n$ modes, only the compliance contributions (i.e., the contributions of the nonrigid body modes evaluated at $\omega = 0$) are taken into account. The sum of compliances represents the residual flexibility part associated with the deformation of the stage model (60) and denoted by $\mathcal{P}_{\text{res}}$.

Given the control context of **Figure 1** with the stage system in Equation 1, the following stage control problems can be formulated (61).

Definition 1 (servo tracking problem). We define the servo tracking problem as finding the feedforward controller $\mathcal{C}_{\text{ff}}$ that, for any $e(0) = e_0$ and in the absence of disturbances, gives

$$\lim_{t \rightarrow \infty} e(t) \stackrel{\text{def}}{=} r(t) - y(t) = 0. \quad 2.$$

By application of the final value theorem, Equation 2 can be written as

$$\lim_{s \rightarrow 0} se(s) = sS(s)(1 - \mathcal{P}(s)\mathcal{C}_{\text{ff}}(s))r(s) = 0, \quad 3.$$

where $S(s) = (1 + \mathcal{P}(s)\mathcal{C}_{\text{fb}}(s))^{-1}$ is the (stable) transfer function known as the sensitivity function. For application of the final value theorem, we have that the Laplace transform of $e(t)$, denoted as $e(s)$, has a single pole at the origin [from $r(s)$ in Reference 57] and its remaining poles in the open left half-plane (62). From Equation 3, the nominal feedforward controller for stages is often chosen as

$$\mathcal{C}_{\text{ff}}(s) \stackrel{\text{def}}{=} \mathcal{P}^{-1}(s) \approx \mathcal{P}_{\text{rb}}^{-1}(s), \quad 4.$$

i.e., a rigid body approximation of the inverse of the plant. As the feedforward controller in Equation 4 only approximates the true plant inverse, which in practice is always the case, a residual in tracking error remains. Apart from the mismatch in feedforward compensation, error residuals in stage control systems also result from disturbances (e.g., d_u , d_y , and n_y) acting on the loop. Error residuals play a role in the disturbance rejection problem.

Definition 2 (disturbance rejection problem). We define the disturbance rejection problem as finding a stabilizing feedback controller $\mathcal{C}_{\text{fb}}$ that minimizes the causal closed-loop error signal e in view of disturbances, or

$$\arg \min_{\mathcal{C}_{\text{fb}}(s) \in \mathbb{C}} \underbrace{\frac{1}{2\pi} \int_{-\infty}^{\infty} |e(j\omega)|^2 d\omega}_{\text{Parseval's theorem}} = \int_0^{\infty} |e(t)|^2 dt \quad \text{with } s = j\omega. \quad 5.$$

For the disturbance rejection problem—similar to the servo tracking problem—there often exists no feedback controller C_{fb} that leads to rejection at all frequencies. This is because of area constraints, i.e., design limitations on the sensitivity function S such as Bode's sensitivity integral (63), the latter of which is often referred to as the waterbed effect. As the feedforward controller C_{ff} does not affect stability properties of the closed loop, it does not play a role in the sensitivity function S and therefore can be designed independently from the feedback controller. That said, the feedforward controller does correlate with the error e in Equation 5, while the feedback controller correlates with the error e in Equation 3. In this regard, the error e of a short-stroke wafer stage is brought from the micrometer level with feedback only to the (sub)nanometer level by adding feedforward control, which validates the independent design of the feedback and the feedforward controller.

To improve the stage control design in view of both the servo tracking problem and the disturbance rejection problem, we consider in this review five nonlinear integrators next to a linear integrator in an otherwise linear stage control design. The linear control design reflects current industrial standards on stage control, namely a PID-like controller, a second-order low-pass filter, and a few notch filters (64). In the Laplace domain, the nominal linear controller is given by

$$C_{fb}(s) = \underbrace{k_p \left(1 + \frac{\omega_i}{s}\right)}_{\text{PI controller}} \cdot \underbrace{\frac{s/\omega_{lag} + 1}{s/\omega_{lead} + 1}}_{\text{Lead-lag filter}} \cdot \underbrace{\frac{\omega_{lp}^2}{s^2 + 2\beta\omega_{lp}s + \omega_{lp}^2}}_{\text{Low-pass filter}} \cdot \underbrace{\prod_{j=1}^2 \frac{\omega_{p,j}^2 s^2 + 2\beta_{z,j}\omega_{z,j}s + \omega_{z,j}^2}{\omega_{z,j}^2 s^2 + 2\beta_{p,j}\omega_{p,j}s + \omega_{p,j}^2}}_{\text{Notch filters}}. \quad 6.$$

The controller parameters of the nominal stage controller C_{fb} are given by the loop gain $k_p = 1.95 \times 10^6$ (in N/m), integrator frequency $\omega_i = 2\pi \cdot 34.4$ (in rad/s), lead frequency $\omega_{lead} = 2\pi \cdot 40.9$ (in rad/s), lag frequency $\omega_{lag} = 2\pi \cdot 2,000$ (in rad/s), low-pass frequency $\omega_{lp} = 2\pi \cdot 2,000$ (in rad/s), and dimensionless damping $\beta = 0.40$. For the notch filters, we have the notch pole frequencies $\omega_{p,1} = 2\pi \cdot 697.1$ and $\omega_{p,2} = 2\pi \cdot 294.3$ (in rad/s), corresponding damping coefficients $\beta_{p,1} = 0.05$ and $\beta_{p,2} = 0.05$, notch zero frequencies $\omega_{z,1} = 2\pi \cdot 505.5$ and $\omega_{z,2} = 2\pi \cdot 1,001.9$ (in rad/s), and corresponding damping coefficients $\beta_{z,1} = 0.006$ and $\beta_{z,2} = 0.05$. The nonlinear integrators reviewed in the next section are tested against the linear integrator in the PI controller part of Equation 6, i.e., ω_i/s .

3. NONLINEAR INTEGRATORS

This section discusses the following nonlinear integrators: the Clegg integrator, a CgLP integrator, the HIGS integrator, a variable gain integrator, and a split-path integrator.

3.1. Clegg Integrator

The Clegg integrator, first introduced by Clegg (13) in 1958, is defined as

$$\text{Clegg integrator} = \begin{cases} \dot{x}(t) = \omega_i^* e(t) & \text{if } e(t) \neq 0, \\ x(t^+) = 0 & \text{if } e(t) = 0, \\ u(t) = x(t), \end{cases} \quad 7.$$

with input $e(t)$, state $x(t)$, and output $u(t)$ at time t ; for the purpose of comparison, ω_i^* is chosen as the normalized integrator gain $\omega_i^* := \omega_i/|1 + 4j/\pi| \approx \omega_i/1.619$, where ω_i is the integrator frequency of the nominal controller from Equation 6. As soon as the input equals zero, i.e., $e(t) = 0$, the state $x(t)$ is given a reset to zero, which is denoted by $x(t^+) = 0$. The time-domain behavior of the Clegg integrator to input $e(t) = \hat{e} \sin(\omega t)$ is shown in **Figure 2**.

It can be seen that the Clegg integrator has sign equivalence between input $e(t)$ and output $u(t)$ as $e(t)u(t) \geq 0$ for all t . It cannot keep a nonzero integrator buffer $u(t) = x(t) \neq 0$ if $e(t) = 0$ for

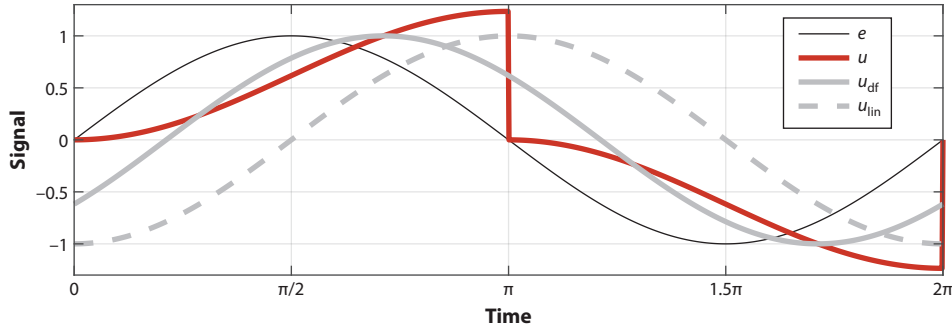


Figure 2

Time series response of the Clegg integrator for $e(t) = \hat{e} \sin(\omega t)$ with $\hat{e} = 1$; u , u_{df} , and u_{lin} are scaled with ω/ω_i , and $\omega = 2\pi \cdot 50$ rad/s.

arbitrary finite time $t > 0$, as in that case $u(t) = x(t) = 0$. Equation 7 represents an impulsive differential equation with a discontinuous output $u(t)$ —as shown in **Figure 2**—which gives strong excitation of stage dynamics and in practice cannot be easily produced by the stage actuator–amplifier combination. Its complexity is fairly low given the one-dimensional state variable $x(t) \in \mathbb{R}$. The Clegg integrator is homogeneous and has scaling invariance because $\alpha e(t) \Rightarrow \alpha u(t)$ for any constant α —i.e., the output $u(t)$ scales linearly with the input $e(t)$.

Using a Fourier series, we can expand the periodic output $u(t)$ from **Figure 2** as

$$u(\tau) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \{a_{2n-1} \sin((2n-1)\tau) + b_{2n-1} \cos((2n-1)\tau)\}, \quad \tau = \omega t. \quad 8.$$

By considering only the components related to $n = 1$, we obtain the approximate solution, as indicated in **Figure 2** by u_{df} , which is known as the solution of the first-order describing function. Describing functions, though approximations, offer indispensable tools for control engineers as they inherit desirable properties from the frequency response method (14). The first-order describing function of the Clegg integrator is defined as

$$\mathcal{N}(\omega) \stackrel{\text{def}}{=} \frac{a_1 + jb_1}{\hat{e}}, \quad 9.$$

where a_1 and b_1 are the Fourier coefficients that follow from the solution u of Equation 7, or

$$u(\tau) = x(\tau) = -\frac{\omega_i^* \hat{e}}{\omega} \left(\cos(\tau) - \text{sign}\{\sin(\tau)\} \right) \quad \text{with } x(0) = 0, \quad 10.$$

which gives

$$a_1 = \frac{1}{\pi} \int_{-\pi}^{\pi} u(\tau) \sin(\tau) d\tau = \frac{4\omega_i^* \hat{e}}{\pi \omega}, \quad b_1 = \frac{1}{\pi} \int_{-\pi}^{\pi} u(\tau) \cos(\tau) d\tau = -\frac{\omega_i^* \hat{e}}{\omega}. \quad 11.$$

These coefficients¹ render the describing function as

$$\mathcal{N}(\omega) := \frac{\omega_i^*}{j\omega} \left\{ 1 + \frac{4j}{\pi} \right\} = \frac{\omega_i}{j\omega} \left\{ \frac{1 + 4j/\pi}{|1 + 4j/\pi|} \right\}, \quad 12.$$

i.e., the frequency response of a simple integrator multiplied with a complex number. The amplitude of the complex number equals $|1 + 4j/\pi| \approx 1.619$, whereas the angle is given by

¹For odd nonlinearities, i.e., all nonlinearities considered in this review, $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} u(\tau) d\tau = 0$.

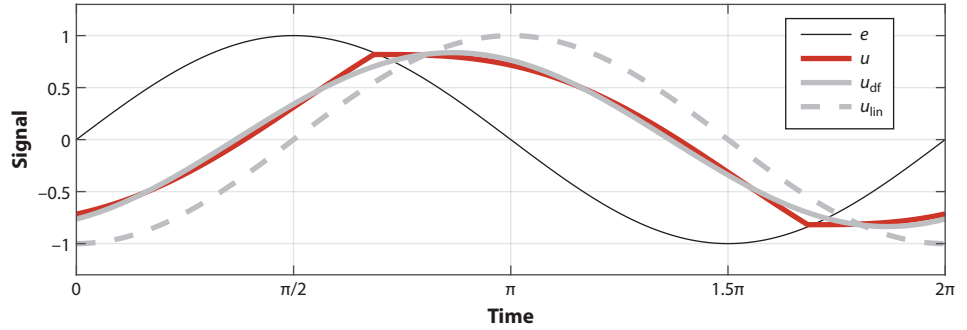


Figure 3

Time series response of the CgLP integrator for $e(t) = \hat{e} \sin(\omega t)$ with $\hat{e} = 1$; u , u_{df} , and u_{lin} are scaled with ω/ω_i , $\omega = 2\pi \cdot 50$ rad/s, and $\omega_r = 2\pi \cdot 20$ rad/s.

$\arctan(4/\pi) \times 180/\pi \approx 51.854^\circ$. As a result, the Clegg integrator's describing function normalized with $|1 + 4j/\pi|$ has magnitude characteristics corresponding to those of a linear integrator with gain ω_i . However, it has a phase of $-90^\circ + 51.854^\circ = -38.146^\circ$ instead of the -90° corresponding to the phase of a linear integrator. The latter is indicated in **Figure 2** by u_{lin} .

3.2. CgLP Integrator

The CgLP integrator—which is based on Horowitz & Rosenbaum's (18) idea of keeping the gain constant while producing a phase lead, and was later reintroduced by Saikumar et al. (29) in 2019—is centered around the FORE:

$$\text{CgLP integrator} = \begin{cases} \text{lead filter: } \bar{e}(t) = \frac{1}{\sqrt{1+4^2/\pi^2}} (\omega_r^* e(t) + \dot{e}(t)), \\ \text{FORE: } \begin{cases} \dot{x}_1(t) = -\omega_r x_1(t) + \omega_r \bar{e}(t) & \text{if } \bar{e}(t) \neq 0, \\ x_1(t^+) = 0 & \text{if } \bar{e}(t) = 0, \\ \bar{u}(t) = x_1(t), \end{cases} \\ \text{linear integrator: } \begin{cases} \dot{x}_2(t) = \frac{\omega_i}{\omega_r} \bar{u}(t), \\ u(t) = x_2(t). \end{cases} \end{cases} \quad 13.$$

In Equation 13, $e(t)$ is the input to a matched lead filter (29), whose output $\bar{e}(t)$ is input to the FORE. Subsequently, the output of the FORE, which is given by $\bar{u}(t)$, is input to a linear integrator, which provides the CgLP integrator output $u(t)$. The other variables are the state $x_1(t)$ of the FORE and the state $x_2(t)$ of the linear integrator at time t ; the pole frequency $\omega_r^* = \alpha \omega_r > 0$, which relates to the pole frequency ω_r of the base linear part of the FORE; the design parameter $\alpha = \sqrt{1 + 4^2/\pi^2}$; and the gain of the linear integrator ω_i .

The FORE has sign equivalence between its input $\bar{e}(t)$ and its output $\bar{u}(t)$ as $\bar{e}(t)\bar{u}(t) \geq 0$ for all t . Sign equivalence does not occur between input $e(t)$ and output $u(t)$ for all t , as can be seen in **Figure 3**. The FORE also has scaling invariance as $\alpha \bar{e}(t) \Rightarrow \alpha \bar{u}(t)$ for any constant α . Similar to the Clegg integrator, the FORE has zero output $\bar{u}(t) = x_1(t) = 0$ when the inputs $e(t), \bar{e}(t) \rightarrow 0$ for (finite) time t . In this case, state x_2 of the CgLP integrator can still provide a nonzero output $u(t) = x_2(t) \neq 0$ (e.g., to lift a stage). As $\bar{u}(t)$ is discontinuous, $u(t)$ is continuous but nonsmooth, which is shown in **Figure 3** and may still cause excitation of stage dynamics. It also requires stronger specifications from the actuator–amplifier combination. In terms of the complexity of the CgLP integrator, a two-dimensional state vector $x(t) = [x_1(t), x_2(t)] \in \mathbb{R}^2$ is used.

In the frequency domain, the describing function of the CgLP integrator follows from the solution $\bar{u}(\tau)$ of Equation 13 for $\bar{e}(t) = \hat{e} \sin(\tau)$, which on the interval $0 \leq \tau < \pi$ with $\tau = \omega t$ and

$x_1(0) = 0$ is given by

$$\bar{u}(\tau) = x_1(\tau) = \frac{\omega_r \hat{e}}{\omega^2 + \omega_r^2} \left(\omega \exp\left(-\frac{\omega_r \tau}{\omega}\right) - \omega \cos(\tau) + \omega_r \sin(\tau) \right) \text{ for } 0 \leq \tau < \pi. \quad 14.$$

After some algebra, this leads to the following Fourier coefficients:

$$\begin{aligned} a_1 &= \frac{2}{\pi} \int_0^\pi u(\tau) \sin(\tau) d\tau = \frac{2\omega_r \hat{e}}{\pi \{\omega^2 + \omega_r^2\}} \left\{ \frac{\omega^3}{\omega^2 + \omega_r^2} \left\{ 1 + \exp\left(-\frac{\omega_r \pi}{\omega}\right) \right\} + \frac{\omega_r \pi}{2} \right\}, \\ b_1 &= \frac{2}{\pi} \int_0^\pi u(\tau) \cos(\tau) d\tau = \frac{2\omega_r \hat{e}}{\pi \{\omega^2 + \omega_r^2\}} \left\{ \frac{\omega^2 \omega_r}{\omega^2 + \omega_r^2} \left\{ 1 + \exp\left(-\frac{\omega_r \pi}{\omega}\right) \right\} - \frac{\omega \pi}{2} \right\}. \end{aligned} \quad 15.$$

Substituting these coefficients into Equation 9 and using the frequency response functions of the lead filter and the integrator from Equation 13 yields the following first-order describing function:

$$\begin{aligned} \mathcal{N}(\omega) &:= \frac{\omega_i}{j\omega} \cdot \frac{\omega_r^* + j\omega}{\omega_r + j\omega} \cdot \left(1 + j \frac{2\omega^2(1 + e^{-\omega_r \pi/\omega})}{\pi(\omega_r^2 + \omega^2)} \right) \cdot \frac{1}{\sqrt{1 + 4^2/\pi^2}} \\ &\approx \frac{\omega_i}{j\omega} \left\{ \frac{1 + 4j/\pi}{|1 + 4j/\pi|} \right\} \text{ if } \omega \gg \omega_r \text{ or if } \omega \rightarrow 0. \end{aligned} \quad 16.$$

The describing function of the CgLp integrator approximately shows the magnitude characteristics of a linear integrator with reduced phase lag going from -90° for $\omega \rightarrow 0$ to $\approx -38.146^\circ$ for $\omega \gg \omega_r$. In **Figure 3**, its solution u_{dlf} is shown with respect to the solution of a linear integrator u_{lin} . Normalization is done with the constant $|1 + 4j/\pi| \approx 1.619$. In the describing function, gain cannot be tuned fully independently from phase, as shown by the magnitude drop of u_{dlf} with respect to u_{lin} . Karbasizadeh & Hosseinia (65) proposed a filtered version of the reset path that utilizes local phase advantages while having a less strong performance impact from high-frequency resets due to gain loss.

3.3. HIGS Integrator

The HIGS integrator was first introduced by van den Eijnden et al. (26) in 2020. It consists of a series connection of a dedicated lead filter, a HIGS element, and a simple integrator. The HIGS integrator is defined as

$$\text{HIGS integrator} = \begin{cases} \text{lead filter: } \bar{e}(t) = e(t) + \frac{k_h \omega_h^{-1}}{\sqrt{1 + 4^2/\pi^2}} \dot{e}(t), \\ \text{HIGS element: } \begin{cases} \dot{x}_1(t) = \omega_h \bar{e}(t) & \text{if } \xi(t) \in \mathbb{F}_{\text{int}}, \\ x_1(t) = k_h \bar{e}(t) & \text{if } \xi(t) \in \mathbb{F}_{\text{gain}}, \\ \bar{u}(t) = x_1(t), \end{cases} \\ \text{linear integrator: } \begin{cases} \dot{x}_2(t) = \omega_i \bar{u}(t), \\ u(t) = x_2(t). \end{cases} \end{cases} \quad 17.$$

In Equation 17, $e(t)$ and its derivative $\dot{e}(t)$ are input to the lead filter. Its output $\bar{e}(t)$ is input to the HIGS element. The output of the HIGS element $\bar{u}(t)$ is input to a linear integrator, which provides the HIGS integrator's output $u(t)$. The other variables are the state $x_1(t)$ of the HIGS element and the state $x_2(t)$ of the linear integrator at time t , the integrator gain $\omega_h > 0$ and proportional gain

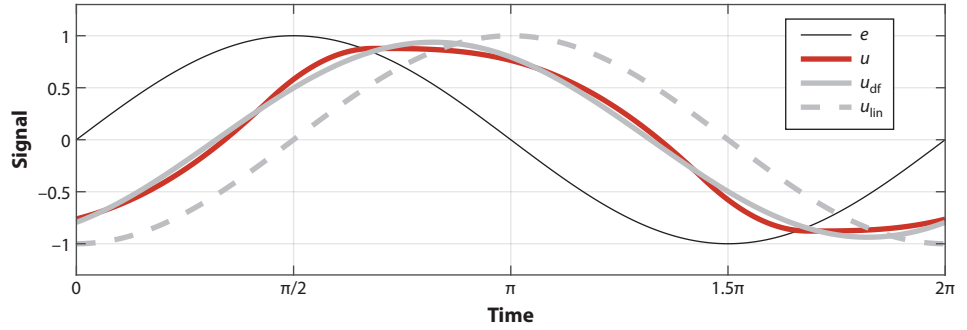


Figure 4

Time series response of the HIGS integrator for $e(t) = \hat{e} \sin(\omega t)$ with $\hat{e} = 1$; u , u_{df} , and u_{lin} are scaled with ω/ω_i , $\omega = 2\pi \cdot 50$ rad/s, $\omega_h = 2\pi \cdot 20$ rad/s, and $k_h = 1$ N/m.

$k_h > 0$ of the HIGS element, and the integrator gain ω_i of the linear integrator. In the HIGS element, the decision is made to switch from an integrator, which is the preferred mode of operation, to a proportional gain and vice versa so as to mitigate the effect of phase lag. The decision variable $\xi(t) = (\bar{e}(t), \dot{\bar{e}}(t), \bar{u}(t))$ is used to decide the mode of operation: $\mathbb{F}_{\text{int}}$ refers to the integrator mode, in which the integrator of the HIGS element gives the output $\bar{u}(t)$, and $\mathbb{F}_{\text{gain}}$ refers to the gain mode, in which the proportional gain term determines the output $\bar{u}(t)$. Note that for the transition to gain mode (or back to integrator mode), we no longer have impulsive behavior as in the Clegg integrator and the FORE. Therefore, impulsive dynamics of the form $x_1(t^+)$ are absent.

Definition 3 (HIGS modes). The integrator mode and the gain mode are respectively defined as

$$\mathbb{F}_{\text{int}} \stackrel{\text{def}}{=} \left\{ \xi \in \mathbb{R}^3 \mid (\bar{e}, \bar{u}) \in \mathbb{F} \right\} \setminus \mathbb{F}_{\text{gain}} \text{ with } \mathbb{F} \stackrel{\text{def}}{=} \left\{ (\bar{e}, \bar{u}) \in \mathbb{R}^2 \mid \bar{e}(t)\bar{u}(t) \geq \frac{1}{k_h} \bar{u}^2(t) \right\}, \quad 18.$$

$$\mathbb{F}_{\text{gain}} \stackrel{\text{def}}{=} \left\{ \xi \in \mathbb{R}^3 \mid (\bar{e}, \bar{u}) \in \mathbb{F} \wedge \bar{u}(t) = k_h \bar{e}(t) \wedge \omega_b \bar{e}^2(t) > k_h \dot{\bar{e}}(t) \bar{e}(t) \right\}. \quad 19.$$

In integrator mode, the input–output pair $(\bar{e}, \bar{u})$ lies primarily within the sector defined by $k_h \bar{e}(t)\bar{u}(t) \geq \bar{u}^2(t)$. In gain mode, $(\bar{e}, \bar{u})$ lies on the sector boundary where $\bar{u}(t) = k_h \bar{e}(t)$. Once in gain mode, HIGS can return to integrator mode only if its vector field points toward the interior of the sector (for more details, see 24, 38, 41). The HIGS element has sign equivalence between its input $\bar{e}(t)$ and its output $\bar{u}(t)$ as $\bar{e}(t)\bar{u}(t) \geq 0$ for all t . Sign equivalence does not apply for input $e(t)$ and output $u(t)$, as shown in **Figure 4**. The HIGS element also has scaling invariance, as $\alpha \bar{e}(t) \Rightarrow \alpha \bar{u}(t)$ for any constant α . Similar to the Clegg integrator, the HIGS element has zero output $\bar{u}(t) = x_1(t) = 0$ in steady state when its input $\bar{e}(t) \rightarrow 0$ for any (finite) time t . For the HIGS integrator, state x_2 can provide a nonzero output $u(t) = x_2(t) \neq 0$ for zero input $e(t) = 0$. This output is continuous and smooth, which gives limited excitation of stage dynamics and puts fewer demands on the actuator–amplifier combination. In terms of complexity, the HIGS integrator uses a two-dimensional state vector $x(t) = [x_1(t), x_2(t)] \in \mathbb{R}^2$ together with switching logic.²

²Issues associated with impulsive systems, like chattering or Zeno behavior, are less relevant to piecewise linear systems like HIGS. Also, stability guarantees for so-called Krasovskii solutions to the HIGS system ensure robustness with respect to small state perturbations, as addressed in Reference 41.

In the frequency domain, the describing function of the HIGS element follows from the solution $\bar{u}(\tau)$ of Equation 17 for $\bar{e}(t) = \hat{e} \sin(\tau)$ with $\tau = \omega t$, which is given by

$$\bar{u}(\tau) = x_1(\tau) = \begin{cases} -\frac{\omega_h \hat{e}}{\omega} (\cos(\tau) - 1) & \text{if } 0 \leq \tau < \gamma, \\ \frac{k_h \hat{e}}{\omega} \sin(\tau) & \text{if } \gamma \leq \tau < \pi, \\ -\frac{\omega_h \hat{e}}{\omega} (\cos(\tau) + 1) & \text{if } \pi \leq \tau < \pi + \gamma, \\ \frac{k_h \hat{e}}{\omega} \sin(\tau) & \text{if } \pi + \gamma \leq \tau < 2\pi, \text{ with } x_1(0) = 0, \end{cases} \quad 20.$$

and $\gamma = 2 \arctan\left(\frac{\omega k_h}{\omega_h}\right)$, which leads to the following Fourier coefficients:

$$\begin{aligned} a_1 &= \frac{1}{\pi} \int_{-\pi}^{\pi} u(\tau) \sin(\tau) d\tau = \frac{\hat{e}}{\pi} \left\{ \frac{\omega_b}{2\omega} (\cos(2\gamma) - 4 \cos(\gamma) + 3) + k_h \left((\pi - \gamma) + \frac{1}{2} \sin(2\gamma) \right) \right\}, \\ b_1 &= \frac{1}{\pi} \int_{-\pi}^{\pi} u(\tau) \cos(\tau) d\tau = \frac{\hat{e}}{2\pi} \left\{ \frac{\omega_b}{\omega} (4 \sin(\gamma) - \sin(2\gamma) - 2\gamma) - k_h (1 - \cos(2\gamma)) \right\}. \end{aligned} \quad 21.$$

These coefficients together with the frequency response functions of the lead filter and the integrator in Equation 17 lead to the first-order describing function of the HIGS integrator:

$$\mathcal{N}(\omega) := \left(\frac{\omega_i}{j\omega} + \frac{k_h \omega_h^{-1} \omega_i}{\sqrt{1 + 4^2/\pi^2}} \right) \frac{a_1 + j b_1}{\hat{e}}. \quad 22.$$

The describing function of the HIGS integrator has the magnitude characteristics of a linear integrator with reduced phase lag beyond the cutoff frequency $\omega_{co} = \omega_h k_h^{-1} \sqrt{1 + 4^2/\pi^2}$. By tuning the ratio ω_h/k_h , the phase in the describing function of the HIGS integrator is designed to have a frequency-dependent transition from -90° , corresponding to a linear integrator, toward -38.146° , corresponding to a Clegg integrator. This transition has a minor impact on the magnitude characteristics. For $\omega_h \ll k_h$, $\gamma \rightarrow \pi$ yields $\mathcal{N}(\omega) \rightarrow (\omega_i/j\omega)(1 + 4j/\pi)/|1 + 4j/\pi|$, i.e., the describing function of the Clegg integrator normalized with the constant $|1 + 4j/\pi| \approx 1.619$. For $\omega_h \gg k_h$, $\gamma \rightarrow 0$ implies $\mathcal{N}(\omega) \rightarrow \omega_i/j\omega$, i.e., the linear integrator from Equation 6. Comparing the solution of the describing function u_{df} with the solution of a linear integrator u_{lin} (see **Figure 4**) clearly shows the phase advantage. Chu et al. (66) studied a generalization called filtered HIGS, in which $\bar{e}$ and $\dot{\bar{e}}$ receive extra filtering regarding the decision variable ξ , i.e., more design freedom.

3.4. Variable Gain Integrator

The variable gain integrator consists of a phase-based variable gain element placed in series with a linear integrator. The phase-based variable gain element was first proposed by Hunnekens et al. (23) in 2016. The variable gain integrator is defined as

$$\text{Variable gain integrator} = \begin{cases} \text{variable gain element: } \begin{cases} \bar{u}(t) = e(t) & \text{if } e(t)\dot{e}(t) > 0, \\ \bar{u}(t) = 0 & \text{if } e(t)\dot{e}(t) \leq 0, \end{cases} \\ \text{linear integrator: } \begin{cases} \dot{x}(t) = \omega_i^* \bar{u}(t), \\ u(t) = x(t), \end{cases} \end{cases} \quad 23.$$

with input $e(t)$, output $\bar{u}(t)$ of the phase-based variable gain element, state $x(t)$ and output $u(t)$ of the variable gain integrator at time t , normalized gain $\omega_i^* = \omega_i/|1/2 + j/\pi|$, and integrator gain $\omega_i > 0$. The product $e(t)\dot{e}(t) = d\{e^2(t)/2\}/dt$ can be interpreted as power in the sense of the derivative of work, i.e., force $ke(t)$ times displacement $e(t)$ with gain $k = 1/2$. To counteract an increase

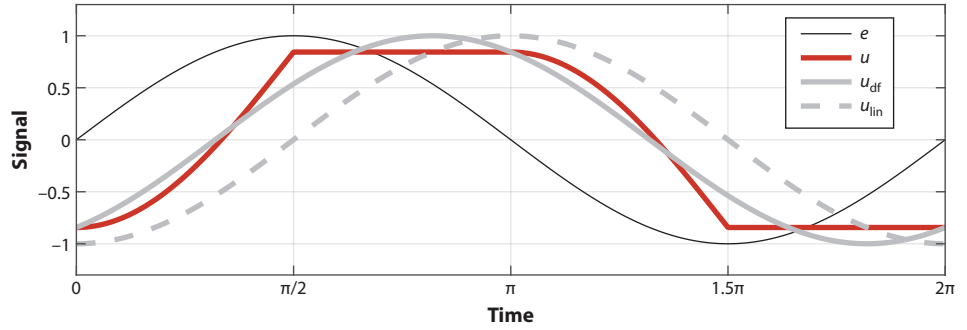


Figure 5

Time series response of the variable gain integrator for $e(t) = \hat{e} \sin(\omega t)$ with $\hat{e} = 1$; u , u_{df} , and u_{lin} are scaled with ω/ω_i , and $\omega = 2\pi \cdot 50$ rad/s.

of power, integration takes place if $e(t)\dot{e}(t) > 0$. If $e(t)\dot{e}(t) \leq 0$, no integration takes place, as power decreases anyway. As can be seen in **Figure 5**, the variable gain integrator has a continuous but nonsmooth output $u(t)$ that gives excitation of stage dynamics and poses stronger demands on the actuator–amplifier combination; the output $\tilde{u}(t)$ is a discontinuous signal. Also, there is no sign equivalence between input $e(t)$ and output $u(t)$ for all t . The integrator can produce a nonzero steady-state output $u(t) = x(t) \neq 0$ for zero input $e(t) = 0$ at arbitrary (finite) time t . Similar to the previous nonlinear elements, the phase-based variable gain element is homogeneous in the sense that $\alpha e(t) \Rightarrow \alpha \tilde{u}(t)$ for any constant α . Regarding complexity, the variable gain integrator has a one-dimensional state vector and no dedicated tuning variable.

For the variable gain integrator, the solution $u(\tau)$ for $e(\tau) = \hat{e} \sin(\tau)$ with $\tau = \omega t$ reads

$$u(\tau) = x(\tau) = \begin{cases} -\frac{\omega_i^* \hat{e}}{\omega} (\cos(\tau) - 1/2) & \text{if } 0 \leq \tau < \pi/2, \\ \frac{\omega_i^* \hat{e}}{2\omega} & \text{if } \pi/2 \leq \tau < \pi, \\ -\frac{\omega_i^* \hat{e}}{\omega} (\cos(\tau) + 1/2) & \text{if } \pi \leq \tau < 3\pi/2, \\ -\frac{\omega_i^* \hat{e}}{2\omega} & \text{if } 3\pi/2 \leq \tau < 2\pi, \end{cases} \quad \text{with } x(0) = -\frac{\omega_i \hat{e}}{2\omega}, \quad 24.$$

which leads to the Fourier coefficients

$$\begin{aligned} a_1 &= \frac{1}{\pi} \int_{-\pi}^{\pi} u(\tau) \sin(\tau) d\tau = \frac{\omega_i^* \hat{e}}{\pi \omega}, \\ b_1 &= \frac{1}{\pi} \int_{-\pi}^{\pi} u(\tau) \cos(\tau) d\tau = -\frac{\omega_i^* \hat{e}}{2\omega}, \end{aligned} \quad 25.$$

and renders the first-order describing function of the variable gain integrator as

$$N(\omega) := \frac{\omega_i}{j\omega} \left(\frac{1/2 + j/\pi}{|1/2 + j/\pi|} \right). \quad 26.$$

Equation 26 shows the magnitude characteristics of a linear integrator with gain ω_i , which is normalized by gain $|1/2 + j/\pi| \approx 0.5927$, and with a reduced phase lag of $\arctan(2/\pi) \times 180/\pi \approx 32.482^\circ$, i.e., a phase of $-90^\circ + 32.482^\circ = -57.518^\circ$. The solution of the describing function u_{df} is shown in **Figure 5** in comparison with a linear integrator u_{lin} , which shows a clear phase lead.

3.5. Split-Path Integrator

The split-path integrator from Karybakas (19), introduced in 1977, uses a differentiator placed before a relay element and in parallel to a rectifier element. The output is found from linear integration. Van Loon et al. (34) reintroduced the split-path integrator in 2016 by placing the linear integrator before the rectifier and relay element, thereby circumventing the need for a differentiator. Sharif et al. (36) generalized the lead filter (or any LTI filter, for that matter) before the relay element to control the switching instances. These split-path integrators all have a discontinuous output, which is undesirable in stage control. To circumvent this, a modified split-path integrator with continuous output is defined as

$$\text{Split-path integrator} = \begin{cases} \text{lead-lag filter: } \begin{cases} \dot{x}_1(t) = -\omega_{\text{lag}}x_1(t) - (1 - \omega_{\text{lag}}/\omega_{\text{lead}})e(t), \\ \bar{e}(t) = -\omega_{\text{lag}}x_1(t) + (\omega_{\text{lag}}/\omega_{\text{lead}})e(t), \end{cases} \\ \text{split-path element: } \bar{u}(t) = \begin{cases} e(t) & \text{if } e(t)\bar{e}(t) \geq 0, \\ -e(t) & \text{if } e(t)\bar{e}(t) < 0, \end{cases} \\ \text{linear integrator: } \begin{cases} \dot{x}_2(t) = \omega_i\bar{u}(t), \\ u(t) = x_2(t), \end{cases} \end{cases} \quad 27.$$

with input $e(t)$, state $x_1(t)$ of a lead-lag filter with $\omega_{\text{lag}} > \omega_{\text{lead}} > 0$, state x_2 of the integrator, filtered error $\bar{e}(t)$ as output of the lead-lag filter, output $\bar{u}(t)$ of the split-path element, and output $u(t)$ of the split-path integrator at time t with integrator gain $\omega_i > 0$. Similar to the variable gain integrator, the product $e(t)\bar{e}(t)$ can be seen as power, in particular if $\bar{e}(t) \rightarrow \dot{e}(t)$. If this power is positive, then the input $e(t)$ is integrated to counteract this power. In contrast to the variable gain integrator, if power becomes negative, then $-e(t)$ is integrated to speed up the rate of change of power toward $e(t)\dot{e}(t) \rightarrow 0$ for $t \rightarrow \infty$. As can be seen in **Figure 6**, the split-path integrator has a continuous but nonsmooth output $u(t)$, while the output $\bar{u}(t)$ of the split-path element is discontinuous. These properties ensure the excitation of the stage dynamics and can be considered actuator unfriendly. There is no sign equivalence between input $e(t)$ and outputs $\bar{u}(t)$ and $u(t)$ for all t . The integrator admits a nonzero steady-state output $u(t) = x_2(t) \neq 0$ for zero input $e(t) = 0$ at arbitrary (finite) time t . Similar to the HIGS element, the split-path element is homogeneous in the sense that $\alpha e(t) \Rightarrow \alpha \bar{u}(t)$ for any constant α . Given the order of the filter in between error $e(t)$ and filtered error $\bar{e}(t)$, the split-path integrator uses a multidimensional state vector.

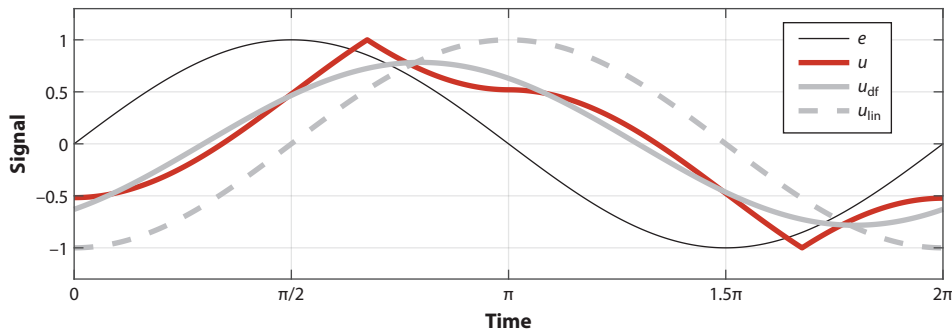


Figure 6

Time series response of the split-path integrator for $e(t) = \hat{e} \sin(\omega t)$ with $\hat{e} = 1$; u , u_{df} , and u_{lin} are scaled with ω/ω_i , $\omega = 2\pi \cdot 50$ rad/s, $\omega_{\text{lag}} = 2\pi \cdot 300$ rad/s, and $\omega_{\text{lead}} = 2\pi \cdot 20$ rad/s.

In the frequency domain, the describing function of the split-path integrator is derived from the solution $\bar{u}(\tau)$ for $e(t) = \hat{e} \sin(\tau)$ with $\tau = \omega t$, which reads

$$\bar{u}(\tau) = \begin{cases} \hat{e} \sin(\tau) & \text{if } 0 \leq \tau < \pi - \eta, \\ -\hat{e} \sin(\tau) & \text{if } \pi - \eta \leq \tau < \pi, \\ \hat{e} \sin(\tau) & \text{if } \pi \leq \tau < 2\pi - \eta, \\ -\hat{e} \sin(\tau) & \text{if } 2\pi - \eta \leq \tau < 2\pi, \end{cases} \quad \eta = \arctan\left(\frac{\omega(\omega_{\text{lag}} - \omega_{\text{lead}})}{\omega^2 + \omega_{\text{lag}}\omega_{\text{lead}}}\right), \quad 28.$$

$0 < \eta \leq \pi/2$, and $\bar{u}(0) = 0$. This gives the following Fourier coefficients:

$$\begin{aligned} a_1 &= \frac{1}{\pi} \int_{-\pi}^{\pi} \bar{u}_2(\tau) \sin(\tau) d\tau = \frac{(\pi - 2\eta)\hat{e}}{\pi} + \frac{\hat{e}}{\pi} \sin(2\eta), \\ b_1 &= \frac{1}{\pi} \int_{-\pi}^{\pi} \bar{u}(\tau) \cos(\tau) d\tau = \frac{\hat{e}}{\pi} - \frac{\hat{e}}{\pi} \cos(2\eta). \end{aligned} \quad 29.$$

These coefficients, together with the frequency response function of the integrator in Equation 27, render the first-order describing function of the split-path integrator as

$$\mathcal{N}(\omega) := \frac{\omega_i}{j\omega} \left(1 + \frac{\sin(2\eta) - 2\eta}{\pi} + j \left(\frac{1 - \cos(2\eta)}{\pi} \right) \right). \quad 30.$$

The describing function of the split-path integrator approximately shows the magnitude characteristics of a linear integrator with gain ω_i but with a reduced phase lag that for the given lead-lag filter can be shaped anywhere between 90° for $\eta = 0$ and 0° for $\eta \rightarrow \pi/2$. **Figure 6** shows the solution of the describing function u_{df} together with the solution of a linear integrator u_{lin} . Similar to the CgLp and HIGS integrators, gain cannot be tuned fully independently from phase, as can be seen from the magnitude drop of u_{df} with respect to u_{lin} .

4. TOOLS FOR STABILITY ANALYSIS

In this section, we present conceptual tools for stability analysis of the stage control system depicted in **Figure 1**, where the linear integrator in the nominal feedback controller C_{fb} from Equation 6 is replaced by one of the five nonlinear integrators discussed in Section 3. Recall that these nonlinear integrators are constructed by combining a nonlinear element N (e.g., a FORE, HIGS, variable gain, or split-path element) with specifically chosen linear filters. By including these linear filters in the remaining linear parts of the system, one can reconstruct the closed-loop control system as the feedback interconnection of a linear system G and a nonlinear element N , as depicted in **Figure 7**.

The linear system G can be any LTI system but here represents the motion stage to be controlled along with all linear elements in the feedback controller. It can be represented in state-space

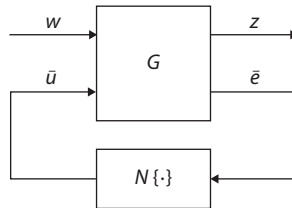


Figure 7

Closed-loop configuration, showing a linear system G in feedback with a nonlinearity $N\{\cdot\}$.

form as

$$G : \begin{cases} \dot{x}_g = A_g x_g + B_g \bar{u} + B_w w, \\ \bar{e} = C_g x_g + D_w w, \\ z = C_z x_g + D_z w, \end{cases} \quad 31.$$

with state vector $x_g(t) \in \mathbb{R}^{n_g}$, external inputs $w(t) \in \mathbb{R}^{n_w}$ containing reference profiles $r(t)$, disturbances $d_u(t)$ and $d_y(t)$, noise signals $n_y(t)$, and performance outputs $z(t) \in \mathbb{R}^{n_z}$ that include control signals and servo tracking errors. The output signal $\bar{e}(t) \in \mathbb{R}$ is fed back to the nonlinearity $N\{\bar{e}(t)\}$, which, in turn, produces the input $\bar{u}(t) \in \mathbb{R}$. Since this connection creates an internal feedback loop, in the upcoming stability analysis the linear transfer function from the input $\bar{u}$ to the output $\bar{e}$ is important. This transfer function is given by

$$G_{\bar{u}\bar{e}}(s) = C_g(sI - A_g)^{-1}B_g = \frac{\mathcal{P}(s)\mathcal{C}(s)}{1 + \mathcal{P}(s)\mathcal{C}(s)} \cdot \mathcal{L}(s) \cdot \mathcal{I}(s), \quad 32.$$

where $\mathcal{C}(s)$ is the nominal controller, and $\mathcal{L}(s)$ and $\mathcal{I}(s)$ are the specific linear pre- and postfilters, respectively, arising in the different nonlinear integrator designs, such as lead filters and integrators.

Based on the characteristics of $N\{\cdot\}$, each of the nonlinear integrators can be categorized into one of three classes: (a) impulsive elements, such as the Clegg and CgLp integrators; (b) piecewise linear elements with a dynamic mode, such as HIGS; and (c) piecewise linear elements with static modes, including variable gain and split-path elements. The different natures of these elements give rise to different closed-loop system models, which all fit the framework of hybrid dynamical systems (67) and require tailored tools for stability analysis. For each of the categories mentioned above, we discuss the specific closed-loop system models and provide analysis tools. The tools are presented from two viewpoints: a time-domain view, leading to numerical tests based on LMIs, and a frequency-domain view, leading to graphical tests.

4.1. Input-to-State Stability of Hybrid Dynamical Systems

Consider a hybrid dynamical system of the form

$$\mathcal{H} : \begin{cases} \dot{x}(t) = f(x(t), w(t)) & \text{if } (x(t), w(t)) \in \mathcal{F}, \\ x^+ = g(x(t)) & \text{if } (x(t), w(t)) \in \mathcal{J}, \end{cases} \quad 33.$$

with state $x(t) \in \mathbb{R}^n$ and inputs $w(t) \in \mathbb{R}^m$, and where f and g are (possibly discontinuous) functions and $\mathcal{F}, \mathcal{J} \subseteq \mathbb{R}^n$ are the jump and flow sets, respectively. We adopt the solution concept for hybrid systems as discussed by Cai & Teel (68, section 2) and assume that, given an initial condition $x(0) = x_0 \in \mathcal{F} \cup \mathcal{J}$ and inputs $w(t)$, solutions to $\mathcal{H}$ exist at least locally³—i.e., the system admits a solution on $[0, T]$ for some $T > 0$. Stability properties of the hybrid system $\mathcal{H}$ are studied in terms of input-to-state stability (70), which is defined as follows.⁴

Definition 4 (input-to-state stability). We call the hybrid system $\mathcal{H}$ input-to-state stable (ISS) if there exist a class $\mathcal{KL}$ function α and a class $\mathcal{K}$ function γ such that for any initial

³We allow for maximal solutions not being complete (e.g., not having solutions with domains that are unbounded in the t direction) in order to separate conditions for the completeness of solutions from conditions for stability (67–69).

⁴For zero inputs $w = 0$, we are interested in the stability of the origin of Equation 33 rather than the stability of a set including the origin.

condition $x(0) = x_0 \in \mathcal{F} \cup \mathcal{J}$ and inputs w satisfying $\sup_{0 \leq \tau \leq t} \|w(\tau)\| < \infty$, all corresponding solutions to $\mathcal{H}$ satisfy

$$\|x(t)\| \leq \alpha(\|x(0)\|, t) + \gamma \left(\sup_{0 \leq \tau \leq t} \|w(\tau)\| \right) \quad 34.$$

for all $t \in [0, T]$, with $T > 0$.

The input-to-state stability property in Definition 4 is assessed for the hybrid system $\mathcal{H}$ by searching for appropriate ISS Lyapunov functions, as formalized in the next result, which is adopted from Cai & Teel (68) and slightly modified to fit the setting described in this article.

Theorem 1. If there exists a locally Lipschitz continuous function $V(x)$ that satisfies

$$c_1 \|x\|^p \leq V(x) \leq c_2 \|x\|^p \quad \text{for all } x \in \mathcal{F} \cup \mathcal{J}, \quad 35a.$$

$$\frac{\partial V(x)}{\partial x} f(x, w) < -c_3 \|x\|^p + c_4 \|w\|^p \quad \text{for all } x \in \mathcal{F}, \quad 35b.$$

$$V(g(x)) - V(x) \leq -c_5 \|x\|^p \quad \text{for all } x \in \mathcal{J}, \quad 35c.$$

where $p \in \mathbb{N}$, $c_i > 0$ for $i = \{1, 2, 3, 4\}$, and $c_5 \geq 0$, then the hybrid system $\mathcal{H}$ in Equation 33 is ISS in the sense of Definition 4. We call V an ISS Lyapunov function.

4.2. Impulsive Systems: Clegg and CgLP Integrators

The Clegg integrator and FORE element discussed in Sections 3.1 and 3.2, respectively, make explicit use of state resets to facilitate improved integrator characteristics. As a result of this resetting behavior, a feedback interconnection of G in Equation 31 and a Clegg integrator or FORE element can be modeled as a hybrid system of the form

$$\mathcal{H}_{\{\text{Clegg}, \text{CgLP}\}} : \begin{cases} \dot{x} = Ax + Bw, & \text{if } \xi \in \mathcal{F} = \{\xi \in \mathbb{R}^m \mid \xi^\top S \xi \geq 0\}, \\ x^+ = Rx, & \text{if } \xi \in \mathcal{J} = \{\xi \in \mathbb{R}^m \mid \xi^\top S \xi \leq 0\}, \\ z = Cx + Dw, \end{cases} \quad 36.$$

in which $x = [x_g^\top, x_r^\top]^\top \in \mathbb{R}^n$ is the augmented state vector containing plant states x_g and reset states x_r , and $\xi = [x^\top, w^\top]^\top \in \mathbb{R}^m$. The specific system matrices of the closed-loop model in Equation 36 with either a Clegg or CgLP integrator are given by, respectively,

$$\text{Clegg} : \left\{ \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] = \left[\begin{array}{cc|c} A_g & B_g & B_w \\ \omega_l^* C_g & 0 & \omega_l^* D_w \\ \hline C_z & 0 & D_z \end{array} \right], R = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}, \text{ and } S = \begin{bmatrix} 0 & C_g^\top & 0 \\ C_g & 0 & D_w \\ \hline 0 & D_w^\top & 0 \end{bmatrix} \right\}, \quad 37.$$

and

$$\text{CgLP} : \left\{ \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] = \left[\begin{array}{cc|c} A_g & B_g & B_w \\ \omega_r C_g & -\omega_r & \omega_r D_w \\ \hline C_z & 0 & D_z \end{array} \right], R = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}, \text{ and } S = \begin{bmatrix} 0 & C_g^\top & 0 \\ C_g & 0 & D_w \\ \hline 0 & D_w^\top & 0 \end{bmatrix} \right\}, \quad 38.$$

where $A_g, B_g, B_w, C_g, C_z, D_z$, and D_w result from the LTI plant G in Equation 31.

4.2.1. Stability tests. To study stability properties of the reset control system $\mathcal{H}_{\{\text{Clegg}, \text{CgLp}\}}$, we aim to verify the existence of a suitable function V that satisfies the conditions in Theorem 1 and thereby certifies the input-to-state stability of the reset control system. For the sake of exposition, we restrict our attention to the specific class of quadratic functions that can be parameterized as $V(x) = x^\top P x$, where P is a symmetric $n \times n$ matrix. There are two approaches for verifying whether a quadratic ISS Lyapunov function exists: Either we search for the function directly by solving a set of LMIs numerically, or we guarantee the existence of such a function implicitly by checking certain frequency-domain conditions. Both the LMI-based (or time-domain-based) test and a frequency-domain test are summarized next without further proofs. For details on the origin of the tests, along with proofs, we direct interested readers to, e.g., References 40 and 47.

Theorem 2 (LMI stability test for reset control systems). Suppose there exists a matrix P and constants $\tau_1, \tau_2 \geq 0$ that satisfy the LMIs

$$P \succ 0, \quad 39a.$$

$$A^\top P + PA + \tau_1 MSM^\top \prec 0, \quad 39b.$$

$$R^\top PR - P - \tau_2 MSM^\top \preceq 0, \quad 39c.$$

with matrices A, R , and S given in Equation 37 for the Clegg integrator and in Equation 38 for the CgLp integrator, and where $M = [I_{n \times n}, 0]$. Then the reset control system $\mathcal{H}_{\{\text{Clegg}, \text{CgLp}\}}$ is ISS in the sense of Definition 4.

Theorem 3 (frequency-domain stability test for reset control systems). Suppose that the LTI system $T(s) = -C(s)G_{\tilde{u}c}(s)/(1 - C(s)G_{\tilde{u}c}(s))$ with $G_{\tilde{u}c}(s)$ given in Equation 32, and where $C(s) = \omega_i^*/s$ for the Clegg integrator and $C(s) = \omega_r/(s + \omega_r)$ for the CgLp integrator, is stable. If there exists a number $\beta \geq 0$ such that the frequency-domain inequality

$$\text{Re} \{T(j\omega)\} - \beta \text{Re} \{C(j\omega)(T(j\omega) - 1)\} > 0 \quad 40.$$

is satisfied for all $\omega \in \mathbb{R} \cup \{\infty\}$, then the reset control system $\mathcal{H}_{\{\text{Clegg}, \text{CgLp}\}}$ is ISS in the sense of Definition 4.

The LMI conditions presented in Equation 39 can be solved efficiently using numerical toolboxes based on semidefinite programming, such as YALMIP (71) and MOSEK (72). Solving the LMIs requires a parametric model of the closed-loop system, which for stage systems may be difficult to obtain in practice with sufficient accuracy. Also, the LMI and frequency-domain tests may be conservative because underlying these tests is the search for a quadratic Lyapunov function. To reduce conservatism in the analysis, LMI conditions based on more flexible piecewise quadratic functions have been developed in works such as Reference 53. In these works, the LMI conditions have been extended to facilitate $\mathcal{L}_2$ and $\mathcal{H}_2$ performance analysis as well. Potentially less conservative frequency-domain stability conditions, tailored to specific reset elements, can be found in, e.g., References 32 and 73.

4.3. Piecewise Linear Systems with Dynamic Modes: HIGS Integrator

The HIGS element presented in Section 3.3 switches between linear integrator and linear gain characteristics. As such, a HIGS-based control system can be modeled as a piecewise linear system of the form

$$\mathcal{H}_{\{\text{HIGS}\}} : \begin{cases} \dot{x} = A_1 x + B_1 \bar{w} & \text{if } x \in \mathcal{F}_1 = \{\xi \in \mathbb{R}^m \mid \xi^\top S_1 \xi \geq 0\}, \\ \dot{x} = A_2 x + B_2 \bar{w} & \text{if } \xi \in \mathcal{F}_2 = \{\xi \in \mathbb{R}^m \mid \xi^\top S_2 \xi < 0 \wedge H\xi = 0\}, \\ z = Cx + D\bar{w}, \end{cases} \quad 41.$$

where $x = [x_g^\top, x_h^\top]^\top \in \mathbb{R}^{n_x}$, $n_x = n_g + 1$, is the state vector containing the plant states and HIGS state x_h ; $\bar{w} = [w^\top, \dot{w}^\top]^\top$ is the extended input vector; and $\xi = [x^\top, \bar{w}^\top]^\top$. The system matrices of the closed-loop model in Equation 41 are given by

$$\text{HIGS : } \left\{ \begin{array}{l} \left[\begin{array}{c|c} A_1 & B_1 \\ \hline C & D \end{array} \right] = \left[\begin{array}{cc|cc} A_g & B_g & B_w & 0 \\ \omega_h C_g & 0 & \omega_h D_w & 0 \\ \hline C_z & 0 & D_z & 0 \end{array} \right], & S_1 = \left[\begin{array}{cc|cc} 0 & k_h C_g^\top & 0 & 0 \\ k_h C_g & -2 & k_h D_w & 0 \\ \hline 0 & k_h D_w & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right], \\ \left[\begin{array}{c|c} A_2 & B_2 \\ \hline C & D \end{array} \right] = \left[\begin{array}{cc|cc} A_g & B_g & B_w & 0 \\ k_h C_g A_g & 0 & k_h C_g B_w & k_h D_w \\ \hline C_z & 0 & D_z & 0 \end{array} \right], & S_2 = \left[\begin{array}{cc|cc} 0 & C_g A_g & 0 & 0 \\ C_g A_g & -\frac{2\omega_h}{k_h^2} & C_g B_w & D_w \\ \hline 0 & C_g B_w & 0 & 0 \\ 0 & D_w & 0 & 0 \end{array} \right], \\ H = [k_h C_g \ -1 \ | \ k_h D_w \ 0], \end{array} \right. \quad 42.$$

where the matrices $A_g, B_g, B_w, C_g, C_z, D_z$, and D_w are given in Equation 31. Note that the system matrices in gain mode (i.e., A_2, B_2 , and S_2) result from explicit differentiation of the algebraic constraint $x_h = k_h \bar{e}$ in Equation 17. As in some cases the inputs $w(t)$ directly enter the dynamics of HIGS, the derivative $\dot{w}(t)$ appears in the closed-loop gain-mode dynamics.

4.3.1. Stability tests. In the same spirit as discussed in the previous section for reset control systems, the input-to-state stability of a HIGS-controlled system $\mathcal{H}_{\text{HIGS}}$ can be verified by certifying the existence of a quadratic ISS Lyapunov function. This results in either a direct LMI-based test or an implicit frequency-domain test, as summarized next. Further details and proofs of these results can be found in, e.g., References 51, 54, and 74.

Theorem 4 (LMI stability test for HIGS-controlled systems). Suppose there exist a matrix P and constants $\tau_1, \tau_2, \tau_3 \geq 0$ that satisfy the LMIs

$$P - \tau_1 M S_1 M^\top \succ 0, \quad 43a.$$

$$A_1^\top P + P A_1 + \tau_2 M S_1 M^\top \prec 0, \quad 43b.$$

$$\Theta^\top (A_2^\top P + P A_2 - \tau_3 M S_2 M^\top) \Theta \prec 0, \quad 43c.$$

with matrices A_i and S_i , $i = \{1, 2\}$, given in Equation 42, and where $M = [I_{n \times n}, 0]$ and $\Theta = [I, -k_h C_g^\top]^\top$. Then the HIGS-controlled system $\mathcal{H}_{\text{HIGS}}$ is ISS in the sense of Definition 4.

Theorem 5 (frequency-domain stability test for HIGS-controlled systems). Suppose the LTI system $T(s) = k_h G_{\bar{u}\bar{e}}(s)/(1 - k_h G_{\bar{u}\bar{e}}(s))$, with $G_{\bar{u}\bar{e}}(s)$ given in Equation 32, is stable. If there exist constants $\lambda \geq 0$ and $k \geq 1$ such that the frequency-domain inequality

$$1 + \text{Re} \left\{ \left(\left(j\omega - \frac{\omega_h}{k_h} \right) T(j\omega) - k \frac{\omega_h}{k_h} \right) C(j\omega) \right\} - \lambda \text{Re} \{ (1 - k - kT(j\omega)) C(j\omega) \} > 0 \quad 44.$$

is satisfied for all $\omega \in \mathbb{R} \cup \{\infty\}$, then the HIGS-controlled system $\mathcal{H}_{\text{HIGS}}$ is ISS in the sense of Definition 4.

Numerical semidefinite programming tools can be used to solve the LMIs in Equation 43 directly (see, e.g., 71, 72). To further relax the LMI conditions and possibly obtain less conservative stability results, similar to reset control systems, the use of piecewise quadratic functions has been considered for HIGS-controlled systems (see, e.g., the results in References 41 and 51). These works have also extended the conditions toward $\mathcal{L}_2$ and $\mathcal{H}_2$ performance analysis. Simplified frequency-domain conditions can be found in References 41 and 54.

4.4. Piecewise Linear Systems with Static Modes: Variable Gain Integrator and Split-Path Integrator

Both the variable gain and split-path elements, as discussed in Sections 3.4 and 3.5, respectively, employ switching between static modes. Switching is dictated by the sign of specific signals, leading to a piecewise linear system representation of the form

$$\mathcal{H}_{\{\text{var.gain, split-path}\}} : \begin{cases} \dot{x} = A_1 x + B_1 w & \text{if } \xi \in \mathcal{F}_1 = \{\xi \in \mathbb{R}^m \mid \xi^\top S \xi \geq 0\}, \\ \dot{x} = A_2 x + B_2 w & \text{if } \xi \in \mathcal{F}_2 = \{\xi \in \mathbb{R}^m \mid \xi^\top S \xi < 0\}, \\ z = Cx + Dw, \end{cases} \quad 45.$$

where $x = x_g$ and $\xi = [x^\top, w^\top, \dot{w}^\top]^\top$ are the state vector and augmented signal vector, respectively, in the case of the variable gain system, and $x = [x_g^\top, x_f^\top]^\top \in \mathbb{R}^{n_x}$, $n_x = n_g + 1$, and $\xi = [x^\top, w^\top]^\top$ are the state vector and augmented signal vector, respectively, of the split-path system, where the state x_f arises from the lead-lag filter in Equation 27. The specific closed-loop system matrices for the variable gain system and split-path system are given by, respectively,

$$\text{Variable gain : } \begin{cases} \left[\begin{array}{c|c} A_i & B_i \\ \hline C & D \end{array} \right] = \left[\begin{array}{c|c} A_g + k_i B_g C_g & B_w + k_i B_g D_w \\ \hline C_z & D_z \end{array} \right], & k_1 = 1, \quad k_2 = 0, \\ S = \left[\begin{array}{c|c|c} C_g^\top C_g A_g + A_g^\top C_g^\top C_g & C_g^\top C_g B_w & C_g^\top D_w \\ \hline B_w^\top C_g^\top C_g & D_w^\top C_g B_w + B_w^\top C_g^\top D_w & D_w^\top D_w \\ \hline D_w^\top C_g & D_w^\top D_w & 0 \end{array} \right], \end{cases} \quad 46.$$

and

$$\text{Split-path : } \begin{cases} \left[\begin{array}{c|c} A_i & B_i \\ \hline C & D \end{array} \right] = \left[\begin{array}{c|c} A_g + k_i B_g C_g & 0 \\ \hline \left(1 - \frac{\omega_{\text{lag}}}{\omega_{\text{lead}}}\right) C_g - \omega_{\text{lag}} & D_w \end{array} \right], & k_1 = 1, \quad k_2 = -1, \\ S = \left[\begin{array}{c|c|c} 2 \frac{\omega_{\text{lag}}}{\omega_{\text{lead}}} C_g^\top C_g & C_g^\top & \frac{\omega_{\text{lag}}}{\omega_{\text{lead}}} C_g^\top D_w \\ \hline C_g & 0 & D_w \\ \hline \frac{\omega_{\text{lag}}}{\omega_{\text{lead}}} D_w^\top C_g & D_w^\top & 2 \frac{\omega_{\text{lag}}}{\omega_{\text{lead}}} D_w^\top D_w \end{array} \right], \end{cases} \quad 47.$$

where A_g, B_g, B_w, C_g, C_z , and D_z result from the augmented plant in Equation 31.

4.4.1. Stability tests. To guarantee that a variable gain or split-path system $\mathcal{H}_{\{\text{var.gain, split-path}\}}$ is ISS, we again resort to finding a quadratic ISS Lyapunov function, either explicitly with LMIs or implicitly through frequency-domain conditions. Both LMI and frequency-domain tests are summarized as follows; further insights and proofs can be found in, e.g., References 34, 35, 75, and 76.

Theorem 6 (LMI stability test for variable gain and split-path systems). Suppose there exist a matrix P and constants $\tau_1, \tau_2 \geq 0$ that satisfy the LMIs

$$P > 0, \quad 48a.$$

$$A_1^\top P + PA_1 + \tau_1 MSM^\top < 0, \quad 48b.$$

$$A_2^\top P + PA_2 - \tau_2 MSM^\top < 0, \quad 48c.$$

with matrices A_1 , A_2 , and S given in Equation 46 for the variable gain system and in Equation 47 for the split-path system, and where $M = [I_{n \times n}, 0]$. Then the system $\mathcal{H}_{\{\text{var.gain, split-path}\}}$ is ISS in the sense of Definition 4.

Theorem 7 (frequency-domain stability test for variable gain and split-path systems). Suppose the LTI system $T(s) = G_{\tilde{u}\tilde{e}}(s)/(1 - G_{\tilde{u}\tilde{e}}(s))$, with $G_{\tilde{u}\tilde{e}}(s)$ given in Equation 32, is stable. If the frequency-domain inequality

$$\frac{1}{k} + \operatorname{Re} \{T(j\omega)\} > 0 \quad 49.$$

with $k = 2$ is satisfied for all $\omega \in \mathbb{R} \cup \{\infty\}$, then $\mathcal{H}_{\{\text{var.gain, split-path}\}}$ is ISS in the sense of Definition 4.

The LMIs in Equation 48 can be solved efficiently using semidefinite programming tools (71, 72). The condition in Equation 49 is equivalent to the classical circle criterion (46). Extensions of the LMIs in Equation 48 toward steady-state analysis in the presence of constant input disturbances can be found in, e.g., References 34–36. For split-path systems, alternative frequency-domain conditions have been proposed by van den Eijnden et al. (75). These conditions are possibly less conservative than the circle-criterion condition provided in Equation 49 but may have as a drawback that they are more cumbersome to verify.

The results of the frequency-domain closed-loop stability tests according to Equations 40, 44, and 49 are shown in **Figure 8**. For the optimized nonlinear integrator designs used in the benchmark examples in Section 5.2, the required conditions to guarantee closed-loop stability are satisfied by verifying straightforward graphical checks. In these checks, the frequency response data of the stage motion system have been used together with data from the feedback controller from Equation 6 and the different nonlinear integrators from Section 3.

5. BENCHMARK EXAMPLES

In this section, we present two benchmark examples: a step response benchmark and a disturbance rejection benchmark. Our interest lies in showcasing the nonlinear integrators from Section 3. Stability is guaranteed for all designs by applying the tools from Section 4.

5.1. Step Response Benchmark

The nonlinear integrators from Section 3 are compared with a linear integrator by means of step responses. In the comparison, the nominal linear controller from Equation 6 is used with all its parameters fixed. The only thing that is changed is the linear integrator part ω_i/s , which is replaced by one of the normalized nonlinear integrators. The nonlinear integrators are designed to have approximately the same magnitude characteristics as the linear integrator but with the controller parameters $\omega_h = \omega_r = \omega_{\text{lead}}$ chosen such that a phase lead of roughly 50° is obtained around the controller bandwidth of 100 Hz. The Clegg integrator and the variable gain integrator have fixed phase advantages and therefore cannot meet the specification of phase.

As shown in the Bode plots in **Figure 9a**, the magnitude characteristics of all integrators are almost identical from a describing function perspective. The split-path, CgLp, and HIGS integrators have magnitude distortions in their describing functions $\mathcal{N}(\omega)$. The Clegg integrator and the variable gain integrator have no such distortion. The phase corresponding to the nonlinear integrators differs significantly from the -90° of the linear integrator. The split-path integrator has the most freedom in creating a (local) phase advantage. The step responses in **Figure 9b** demonstrate that the phase advantages from the describing function analysis in all nonlinear integrators translate to reduced overshoot in comparison with the overshoot of the linear design of $1.623 \mu\text{m}$. The overshoot and the nonlinear integrator parameters are given in **Table 1**. The

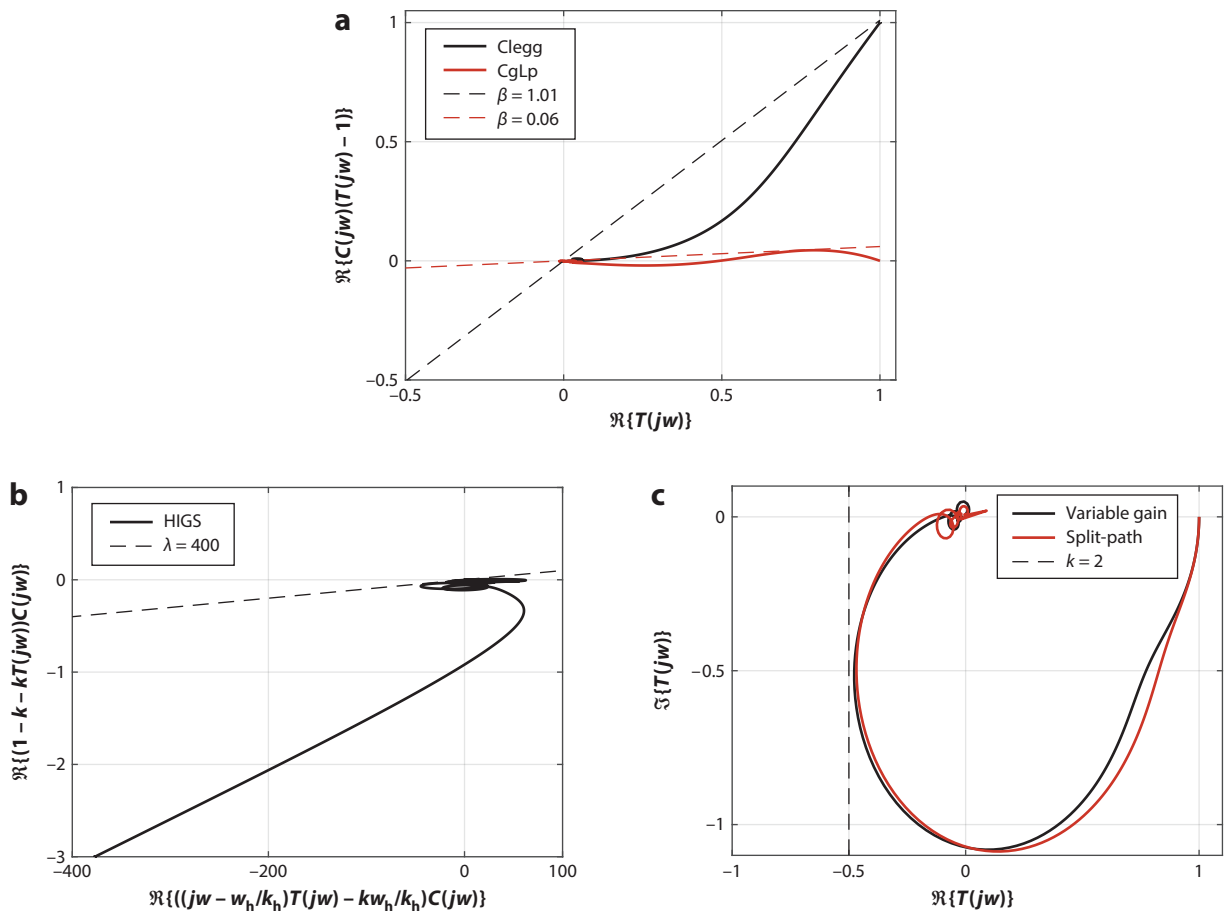


Figure 8

Frequency-domain stability for (a) Equation 40, (b) Equation 44, and (c) Equation 49.

table shows that the control design based on the HIGS integrator gives the biggest reduction in overshoot (52%), whereas the control design based on the variable gain integrator gives the smallest reduction (14%).

5.2. Disturbance Rejection Benchmark

For the disturbance rejection benchmark, a tailored disturbance spectrum is designed that mimics the disturbance spectra encountered during wafer scanning. This benchmark differs from the

Table 1 Nonlinear integrator parameters and overshoot performance

	Clegg	CgLp	HIGS	Variable gain	Split-path
Parameter values ^a	—	$\omega_r = 2\pi \cdot 20$	$\omega_h = 2\pi \cdot 20$ $k_h = 1$	—	$\omega_{\text{lead}} = 2\pi \cdot 20$ $\omega_{\text{lag}} = 2\pi \cdot 300$
Overshoot ^b	1.351 (44%)	1.452 (27%)	1.296 (52%)	1.533 (14%)	1.407 (35%)

^aFrequency ω in rad/s and gain k_h in N/m.

^bOvershoot in μm along with the percentage of reduction compared with a linear design.

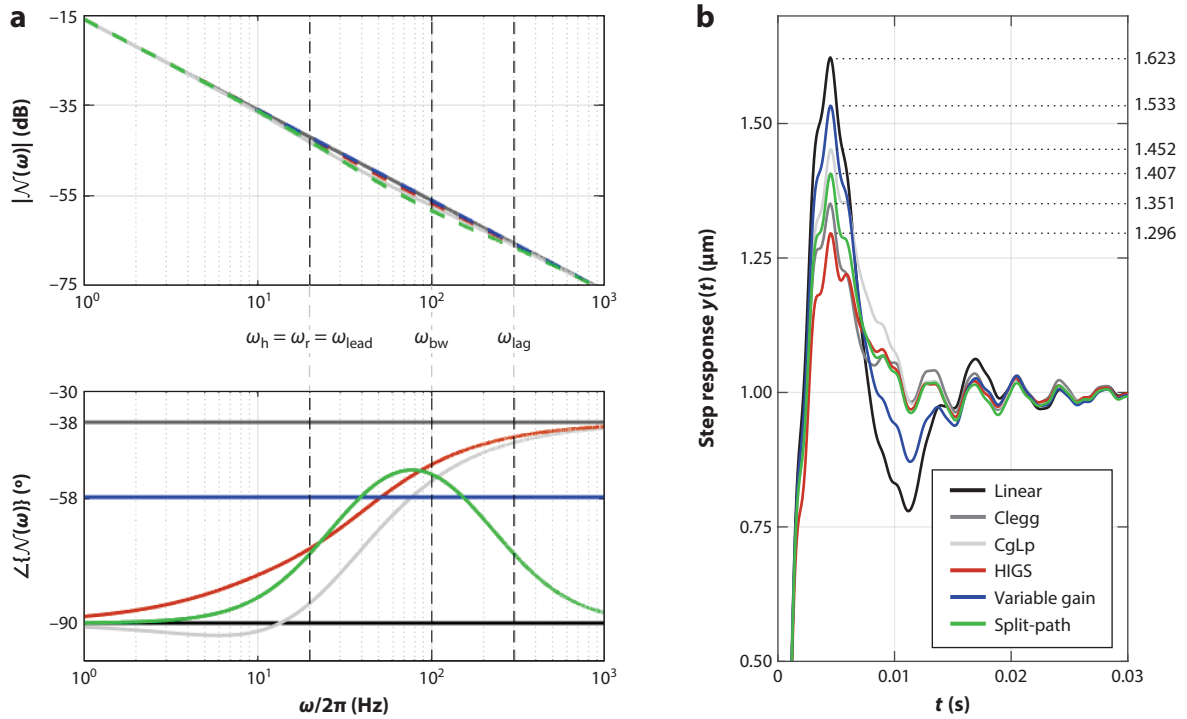


Figure 9

(a) Bode diagrams of describing functions and (b) closed-loop step responses.

step response benchmark in Section 4.3 in the sense that all controller parameters (linear and nonlinear) are subject to optimization. We want to fully exploit the offered design freedom of the nonlinear integrators in showcasing their potential for control.

Given an automated (frequency-domain) tuning procedure, six optimal parameter sets are obtained for the feedback controllers C_b that include the nonlinear integrators. The controllers are shown in **Figure 10a** by using the describing functions from Section 3. From the Bode plots, it can be seen that the magnitude characteristics of the nonlinear feedback controllers all have increased low-frequency gain in comparison with the magnitude characteristics of the optimal linear controller. The highest gain is obtained with the Clegg integrator design. Performance in terms of a cumulative power spectral density analysis is shown in **Figure 10b**. All nonlinear designs (except for the design based on the Clegg integrator) lead to improved disturbance rejection in comparison with the linear control design. The Clegg integrator design, whose results have been scaled to fit the 0.15- μm range, demonstrates the worst performance, showing the difficulty in countering constant gravitational forces while having resets of its integrator buffer.

Disturbance rejection performance is summarized in **Table 2** through the maximum achievable bandwidth ω_{bw} ,⁵ the loop gain k_p , and the integrator frequency ω_i . In addition, the product of loop gain with integrator frequency is used to define the disturbance rejection factor $\eta \stackrel{\text{def}}{=} (k_p \times \omega_i) / (k_p^{\text{lin}} \times \omega_i^{\text{lin}})$, where $k_p^{\text{lin}} = 2.0 \times 10^6$ N/m and $\omega_i^{\text{lin}} = 2\pi \cdot 34.4$ rad/s are the loop gain

⁵ An autotuner is used to maximize the controller bandwidth while satisfying robust stability specifications on closed-loop sensitivity functions. In these sensitivity functions, the nonlinear integrators are approximated by their describing functions (for more details, see 77).

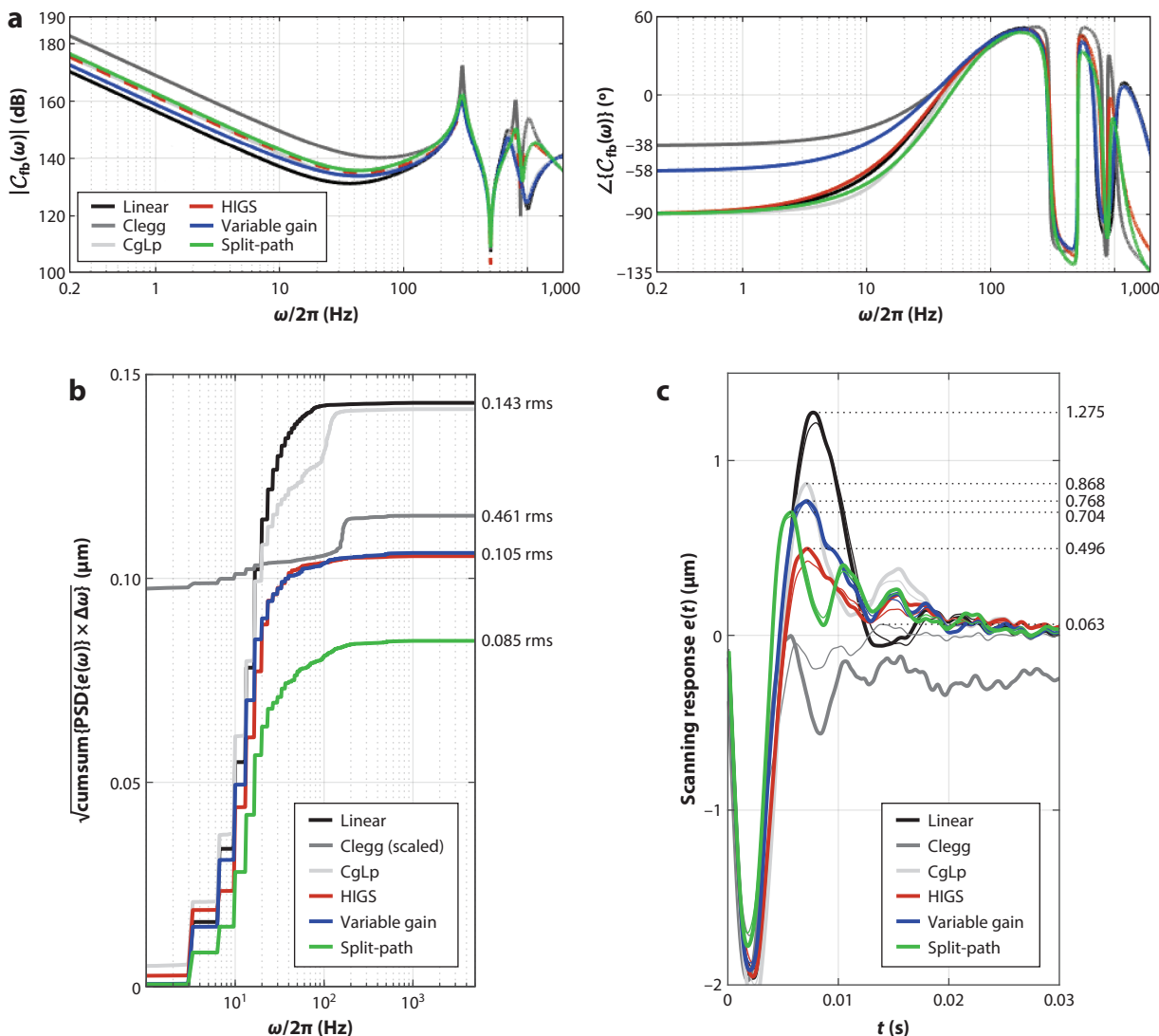


Figure 10

(a) Bode diagrams of the feedback controllers based on describing functions, (b) cumulative power spectral densities, and (c) scanning response measurements. In panel c, thin lines denote simulation results. Abbreviations: PSD, power spectral density; rms, root mean square.

and integrator frequency, respectively, of the optimized linear controller. The design based on the Clegg integrator excels in a bandwidth of 146 Hz (i.e., a 45% increase with respect to the optimal linear control design) and $\eta = 4.1$. The remaining nonlinear designs with $\eta \in [1.5, 1.8]$ indicate a potential improvement with respect to the optimal linear design toward a factor of two. Only the split-path design with $\eta = 1.7$ matches with the measured root-mean-square improvement of $0.143 \mu\text{m}/0.085 \mu\text{m} \approx 1.7$. For HIGS ($\eta = 1.8$) and the variable gain integrator design ($\eta = 1.5$), the disturbance rejection factor is not fully matched by the measured root-mean-square improvement of $0.143 \mu\text{m}/0.105 \mu\text{m} \approx 1.4$. For both the Clegg integrator and the

Table 2 Nonlinear controller design: performance metrics

	Linear	Clegg	CgLp	HIGS	Variable gain	Split-path
$\omega_{bw} (\times 1/2\pi)^a$	101	146	118	118	111	129
$k_p (\times 10^6)^b$	2.0	2.0	1.7	2.0	2.0	2.3
$\omega_i (\times 1/2\pi)^c$	34.4	140.8	72.0	60.7	43.7	59.7
η	1	4.1	1.8	1.8	1.5	1.7
Overshoot ^d	1.275	0.063 ^e	0.868 (31%)	0.496 (61%)	0.704 (45%)	0.768 (40%)

^aBandwidth frequency $\omega_{bw}/2\pi$ in Hz.

^bLoop gain k_p in N/m.

^cIntegrator frequency $\omega_i/2\pi$ in Hz.

^dOvershoot in μm .

^eBased on simulation results.

CgLp integrator, the expected improvements are not confirmed by measurement. These impulsive systems appear to be too sensitive to high-frequency signal content around their bandwidths despite appropriate pre- and postfiltering.⁶ Time series measurements of the scanning responses, shown in **Figure 10c**, demonstrate the occurrence of DC error when using the Clegg integrator. In the absence of constant gravitational forces, the Clegg integrator would have superior performance, as indicated by the time series simulation results (thin curves). Overshoot is significantly reduced for all nonlinear designs. The best reduction is obtained with HIGS and equals $100\% \times (1.275 - 0.496)/1.275 = 61\%$.

6. CONCLUSIONS

Five nonlinear integrator designs have been reviewed and compared with an optimal LTI stage control design. All nonlinear designs consistently demonstrate smaller overshoot, faster settling times, higher controller bandwidths, and higher (low-frequency) controller gains. At the same time, the nonlinear designs have difficulty in beating the simplicity, predictability, robustness, and ease of tuning of the linear control design. Moreover, as the nonlinear designs lack superposition, stability tools at best provide sufficient (not necessary) conditions for closed-loop stability. Lack of superposition leads to frequency-domain tuning being insufficient on its own and necessitates time-domain signal conditioning—such as pre- and postfiltering and/or threshold elements operating on the input signal of the (nonlinear) integrator—in order to avoid unnecessary resets or switching (79) that otherwise will lead to worse performance than predicted on the basis of describing function analysis.

A valuable future asset to the nonlinear control toolbox would be a (multivariable) robust design in the frequency domain (80) in which frequency-domain specifications and loop shaping using describing function analysis are combined with uncertainty models, particularly uncertainty models on the frequency-domain description of the nonlinear integrator (77, 81). Such models should cover the blank spots when using describing functions to, for example, address the problem of gain loss (78). Other engineering approaches in this regard are filtering of the reset/decision branch, as was done with the filtered split-path integrator by van der Maas et al. (35) and Sharif

⁶All nonlinear integrator designs need appropriate pre- and postfiltering for time-domain signal conditioning, which is done with skew notch filters. Without pre- and postfiltering, high-frequency signals will induce unwanted resets or mode switching that may lead to gain loss (78).

et al. (36), with a shaping filter in CgLP by Caporale et al. (82), and with filtered HIGS by Chu et al. (66).

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